Metastasis from Uveal Melanoma after Proton Beam Irradiation

EVANGELOS S. GRAGOUDAS, MD,[1] JOHANNA M. SEDDON, MD,[1] KATHLEEN M. EGAN, MPH,[1] ROBERT J. GLYNN, ScD, PhD,[1] MICHAEL GOITEIN, PhD,[2] JOHN MUNZENRIDER, MD,[2] LYNN VERHEY, PhD,[2] MARCIA URIE, PhD,[2] ANDREAS KOEHLER, AM[3]

Abstract: The incidence of metastasis and prognostic factors for metastasis in 780 consecutive patients with uveal melanomas treated with proton beam irradiation were evaluated. Metastasis developed in 64 patients (8%). The median time from treatment to the diagnosis of metastasis was 2.1 years (range, 3 months to 7.3 years). The liver was primarily involved in 58 (90%) patients. The 5-year cumulative probability of metastasis developing was 20%. Prognostic factors for metastasis developing were quite comparable to those found for patients treated by enucleation and included largest tumor diameter, involvement of the ciliary body, older age, and extrascleral extension. Surgical localization, tumor height, and elevated liver enzymes before treatment were not important factors in the development of metastasis. [Key words: incidence, metastases, prognostic factors, proton beam irradiation, uveal melanoma.] Ophthalmology 95:992–999, 1988

Conservative treatments for uveal melanoma have been used with increased frequency during the last decade.[1–7] Radiation therapy is the most widely used method, and has been proven effective in maintaining vision and saving eyes in many cases, at least during the first few years after treatment.[3–9] Large series of patients who were treated by enucleation showed a high incidence of metastasis 5 to 10 years after surgery.[10–15] Little information is available, however, on the incidence of metastasis among patients treated by irradiation[7,16–19] and followed for a comparable period of time.

Originally received: November 9, 1987.
Revision accepted: February 16, 1988.

[1] Retina Service, Massachusetts Eye and Ear Infirmary, Department of Ophthalmology, Harvard Medical School, Boston.
[2] Department of Radiation Medicine, Massachusetts General Hospital, Boston.
[3] Harvard Cyclotron Laboratory, Cambridge.

Presented at the American Academy of Ophthalmology Annual Meeting, Dallas, November 1987.

Reprint requests to Evangelos S. Gragoudas, MD, Massachusetts Eye and Ear Infirmary, 243 Charles St, Boston, MA 02114.

We have used proton beam irradiation at the Harvard Cyclotron since 1975 and more than 1000 patients have been treated. In this report, the incidence of metastasis and prognostic factors for metastasis in patients with uveal melanomas treated with proton beam irradiation were evaluated in patients treated from July 1975 through December 1985.

SUBJECTS AND METHODS

STUDY POPULATION

All patients who were treated between July 1975 and December 1985 were identified, and follow-up data were analyzed up to December 1986. A total of 808 patients with melanomas of the choroid and/or ciliary body were treated during this interval. No patient had overt metastatic disease before treatment. Only one patient who had metastasis 2 years after irradiation had evidence suggestive of liver metastasis before therapy (abnormal liver function tests and liver scan), but results of liver biopsy were negative. Twenty-eight patients were excluded from analysis, 3 because their tumors were incompletely excised before proton beam irradia-